

Short Communication

Diversification of the VH3-53 immunoglobulin gene segment by somatic hypermutation results in neutralization of SARS-CoV-2 virus variants

Matthias Bruhn¹ , Maureen Obara¹, Abdus Salam¹, Bibiana Costa¹, Annett Ziegler¹, Inken Walzl¹, Andreas Pavlou¹, Markus Hoffmann^{2,3}, Theresa Graalmann^{4,5,6}, Stefan Pöhlmann^{2,3} , Axel Schambach^{7,8,9} and Ulrich Kalinke^{1,9,10}

¹ Institute for Experimental Infection Research, TWINCORE, Centre for Experimental and Clinical Infection Research, a joint venture between the Helmholtz Centre for Infection Research and the Hannover Medical School, Hannover, Germany

² Infection Biology Unit, German Primate Center, Göttingen, Germany

³ Faculty of Biology, Georg-August-University Göttingen, Göttingen, Germany

⁴ Department for Rheumatology and Immunology, Hannover Medical School, Hannover, Germany

⁵ Junior Research Group for Translational Immunology, TWINCORE, Centre for Experimental and Clinical Infection Research, a joint venture between the Helmholtz Centre for Infection Research and the Hannover Medical School, Hannover, Germany

⁶ Biomedical Research in End-Stage and Obstructive Lung Disease Hannover (BREATH), German Center for Lung Research (DZL), Hannover, Germany

⁷ Institute of Experimental Hematology, Hannover Medical School, Hannover, Germany

⁸ Division of Hematology/Oncology, Boston Children's Hospital, Harvard Medical School, Boston, Massachusetts, USA

⁹ German Center for Infection Research (DZIF), partner site Hannover-Braunschweig, Hannover, Germany

¹⁰ Cluster of Excellence RESIST (EXC 2155), Hannover Medical School, Hannover, Germany

COVID-19 induces re-circulating long-lived memory B cells (MBC) that, upon re-encounter with the pathogen, are induced to mount immunoglobulin responses. During convalescence, antibodies are subjected to affinity maturation, which enhances the antibody binding strength and generates new specificities that neutralize virus variants. Here, we performed a single-cell RNA sequencing analysis of spike-specific B cells from a SARS-CoV-2 convalescent subject. After COVID-19 vaccination, matured infection-induced MBC underwent recall and differentiated into plasmablasts. Furthermore, the transcriptomic profiles of newly activated B cells transiently shifted toward the ones of atypical and CXCR3⁺ B cells and several B-cell clonotypes massively expanded. We expressed monoclonal antibodies (mAbs) from all B-cell clones from the largest clonotype that used the VH3-53 gene segment. The *in vitro* analysis revealed that some somatic hypermutations enhanced the neutralization breadth of mAbs in a putatively stochastic manner. Thus, somatic hypermutation of B-cell clonotypes generates an anticipatory memory that can neutralize new virus variants.

Keywords: Affinity maturation · B cells · Clonotype · Diversification · Monoclonal antibody · Neutralization breadth · Somatic hypermutation · VH3-53 · Virus variants



Additional supporting information may be found online in the Supporting Information section at the end of the article.

Correspondence: Prof. Ulrich Kalinke
e-mail: Ulrich.Kalinke@twincore.de

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Introduction

Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) is the causative infectious agent of the worldwide COVID-19 pandemic [1]. While infection- and vaccination-induced neutralizing serum antibodies can prevent infection and/or severe disease, they also drive the emergence of new immune escape virus variants, including highly infective and pathogenic virus variants of concern (VOC), such as the Alpha, Beta, Gamma, Delta, and Omicron variants, posing an extraordinary challenge for the design of effective vaccines and therapeutic monoclonal antibodies (mAbs) [2]. In the case of SARS-CoV-2, certain immunoglobulin variable gene segments such as the VH3-53 gene segment are prone to dominate memory B-cell (MBC) responses against the spike protein and are repeatedly detected in potent neutralizing antibodies [3]. Consequently, virus exposed to polyclonal serum antibodies escapes from neutralization by the insertion of a few mutations in the spike protein that prevent the most dominant antibody classes from binding [4].

An important pillar of the adaptive immune system is the formation of a pool of MBC, which are particularly relevant for the protection against re-infections with the original virus as well as with new virus variants [5]. MBC are detected in the blood early after recovery from COVID-19 and encode for antibodies with close-to-germline variable region sequences [6]. After virus clearance, the SARS-CoV-2 antigen persists for multiple months and spike-specific MBC continue to evolve by the acquisition of somatic hypermutations (SHM) [7]. An antibody response superior in quantity and quality is generated when convalescent individuals are additionally vaccinated [8], or after a breakthrough infection [9]. A similar effect can be observed after booster vaccination [10], indicating that, at least in the case of SARS-CoV-2, repeated antigen contacts enhance the neutralization of virus variants. The mechanism that enhances the neutralization breadth under these conditions is the focus of an ongoing discussion [5]. Pre-existing antibodies may mask immunodominant epitopes to shift the immune response toward subdominant regions, thus diversifying the clonal composition of B-cell responses [5]. Furthermore, pre-existing MBC can be recruited to recall responses [11], thus allowing matured antibodies to further hypermutate and contribute to secondary immune responses. Indeed, antibodies can gain the ability to neutralize VOC, and affinity maturation has a key role in the “anticipation” of viral escape [12]. Furthermore, we demonstrated previously by reversion of SHM that some germline versions of mAbs already potentially neutralized the SARS-CoV-2 wildtype virus and that upon insertion of certain SHM neutralization of VOC was achieved [13]. An improved understanding of this process is of relevance for the design of broadly neutralizing and/or better-targeted vaccines and monoclonal antibodies.

Results and discussion

Memory B cells change their transcriptomic profile after recall

We hypothesized that after antigen re-exposure, mature MBC derived from an earlier antigen exposure may participate in the secondary immune response. To address this, we analyzed blood cells from a convalescent donor who was vaccinated against COVID-19 approximately eleven months post-recovery, and from whom we previously isolated and characterized mAbs, including MB021_C09K, MB027_D06L, and MB027_E04L [13]. The vaccine contained the sequence information of the same ancestral SARS-CoV-2 spike protein as it was present during the original infection. During the vaccine response, the plasma IgG antibodies increased strongly, and the neutralizing activity grew (Fig. S1a). Blood was taken before vaccination and on days 8, 16, 71, and 104 after the vaccination. RBD-specific class-switched B cells were sorted (Fig. S2) and subjected to single-cell RNA-sequencing (scRNA-seq, Fig. 1a). For bioinformatics analysis of the scRNA-seq data, the samples were de-multiplexed based on the presence of individual barcodes in their surface libraries, allowing tracing back each cell to its sampling day with high confidence (Fig. S1b).

The B cells were grouped into CD38⁺ plasmablasts and multiple clusters of CD38[−] MBC (Fig. S3a). The clusters were annotated based on the expression of characteristic marker genes (Fig. S3b and c). Among the four MBC clusters, we identified two recently described populations of the alternative lineages CXCR3⁺ and atypical MBC [14]. The remaining cells were classical MBC, which we divided into infection- and vaccine-induced MBC, depending on whether they were detected in the prevaccination sample or samples collected after the vaccination (Fig. 1b and c). The total number of class-switched RBD-specific B cells in peripheral blood transiently increased during the vaccination. RBD-specific MBC originating from the pre-vaccination sample were almost exclusively found in the infection-induced MBC cluster. The plasmablasts emerged eight days after the vaccination, which was accompanied by an increase in the frequency of CXCR3⁺ MBC. After 16 days, the plasmablast response waned, while a strong shift toward the alternative CXCR3⁺ and atypical MBC lineages was observed. The samples from days 71 and 104 were characterized by a shift back toward classical MBC, and the vaccine-induced MBC fully replaced the infection-induced ones (Fig. 1c). The emergence of atypical (CD21[−]CD27[−]) MBC in recall responses to SARS-CoV-2 was recently confirmed in a substantial cohort [15]. Interestingly, the Janssen Ad26.COV2.S vaccine was reported to elicit a pronounced CXCR3⁺ B-cell response, while mRNA-based vaccines did not [16]. Since the blood donor in our study first received the ChAdOx1-S vaccine, the CXCR3⁺ B-cell response may be a commonality between

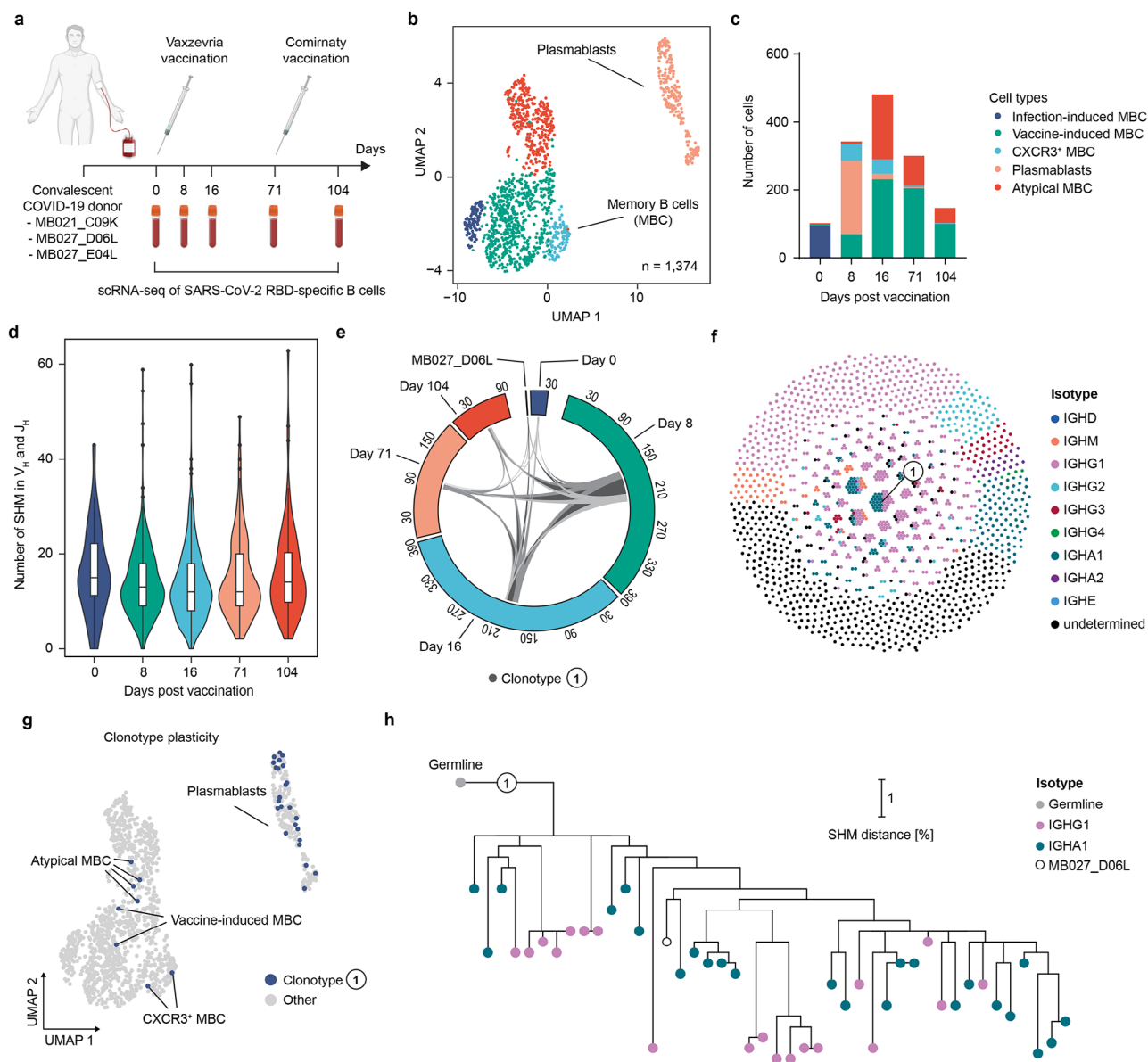


Figure 1. Pre-existing MBC clonotypes diversify and re-engage in COVID-19 vaccine responses. (a) Experimental outline of the study. PBMC from the five time points were barcoded with hashtag oligos and pooled. (b) Unique manifold approximation and projection (UMAP) plot of the quality-filtered dataset. The cell types were identified based on the expression of marker genes (see Fig. S3). (c) The composition of RBD-specific cells over time-resolved for their cell type. The color code in (b) and (c) refers to the legend on the right. (d) Comparison of the number of heavy chain SHM in the five different time points. (e) Chord diagram depicting clonotypes that share at least one cell with the prevaccination sample. The mAb MB027_D06L was added as prevaccination cell. (f) The honeycomb plot was generated using 10X Genomics enclon. Points clustering together indicate the presence of expanded clones. The colors indicate isotypes. The largest clonotype is indicated with the number 1. (g) Location of clonotype 1 in the UMAP clusters. (h) The phylogenetic tree for clonotype 1 visualizes the SHM evolution from a putative germline. The colors indicate isotypes.

adenovirus vector-based vaccines. The construction of a putative trajectory graph revealed that the differentiation from MBC to plasmablasts led through the cluster of CXCR3⁺ MBC, pointing to an involvement of this MBC subpopulation in the plasmablast response (Fig. S1c). Future studies are needed to address which role CXCR3⁺ MBC play in adaptive immune responses.

Pre-existing memory B cells re-enter the vaccine response

To address whether infection-induced MBC participated in vaccine-induced responses, we analyzed the clonal relationships of single B cells based on their V(D)J sequences. Over the course of the vaccination, the extent of SHM did not majorly change

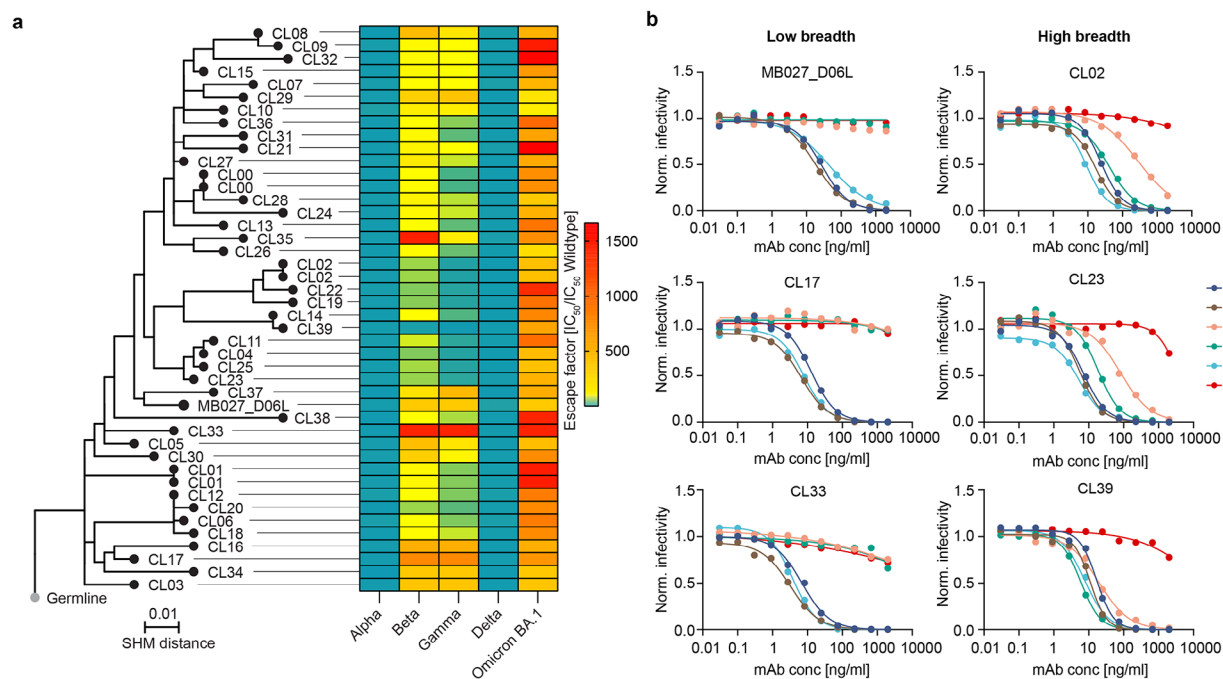


Figure 2. Intraclonotype diversity mediates additional VOC cross-neutralization. (a) The phylogenetic tree of clonotype 1 is shown, including the prevaccination mAb MB027_D06L. All clones were expressed as recombinant IgG1 antibodies and tested for neutralization of VSV pseudotypes displaying spike versions of different SARS-CoV-2 VOCs. An escape factor was calculated for each of the VOCs by normalization of the IC_{50} on the Wildtype strain. For mAbs with identical names and amino acid sequences, the same values are plotted multiple times. (b) Exemplary neutralization curves of some mAbs with low (left) and high (right) neutralization breadth. Nonlinear regression (in case of neutralizing activity) or linear regression (in case of no measurable activity) was performed in GraphPad Prism.

(Fig. 1d), suggesting that pre-existing, already-matured MBC dominated the class-switched B-cell response. Furthermore, we noticed that the largest clonotypes were shared between the prevaccination MBC and the four post-vaccination samples, which proves that they were not induced by the vaccination, but that they were present already before (Fig. 1e). These clonotypes often consisted of mixed isotypes (Fig. 1f). The most expanded clonotype, in the following referred to as clonotype 1, carried the immunodominant VH3-53 gene segment and exerted plasticity, i.e., it occurred in all B-cell subsets and markedly contributed to the plasmablast response (Fig. 1g). The SARS-CoV-2 neutralizing mAb MB027_D06L, which we discovered previously in the same blood donor, also was part of clonotype 1. Substantial diversification occurred within the clonotypes concerning their extent of SHM (Fig. 1h).

B-cell clonotype diversity translates into increased neutralization breadth

To test whether the SHM in clonotype 1 had an influence on the neutralization of VOC, we expressed all members of clonotype 1 as recombinant IgG1 (MB043_CL00 to CL39) and determined their activity against a panel of SARS-CoV-2 VOC (Fig. 2a). Some of these antibodies exhibited a similar neutralizing breadth as the pre-vaccination mAb MB027_D06L, while others had an increased neutralization breadth (Fig. 2b). While many mAbs

were not able to neutralize the Beta, Gamma, and Omicron variants, a large subset of mAbs gained the ability to neutralize these variants by insertion of various SHM (Fig. 3a). Since the effect was most pronounced for the Gamma variant, we performed an amino acid sequence-wide association of antibody variants with the phenotype of cross-neutralization of the Gamma variant and identified three significant positions in the heavy chain (Fig. 3b). Stratification of the antibodies for their genotypes revealed that those carrying a G26E SHM (located in the V_H CDR1) and a 102N SHM (located in the CDR3) powerfully cross-neutralized the Gamma virus variant, while those without these SHM did not (Fig. 3c, full set of comparisons in Fig. S4a). While these SHM are associated strongest with the phenotype, additional beneficial SHM may exist, which did not reach statistical significance in this analysis. The same antibodies that neutralized the Gamma variant also neutralized the Beta variant. This observation makes our conclusion applicable to K417T, but also to K417N, which is present in the Beta and also in the Omicron variant (Fig. S4b). The number of SHM did not significantly correlate with the overall neutralization strength, indicating that an increase in potency is not the major driving force for SHM (Fig. 3d).

In order to identify the responsible mutations on the SARS-CoV-2 RBD, we performed *in silico* homology-controlled antigen docking analysis for all members of clonotype 1. The antibody P5A-1D2 [17] (PDB ID: 7CHO) served as a docking template, which is made up of similar IGHV3-53/IGHJ4 and IGLV1-40

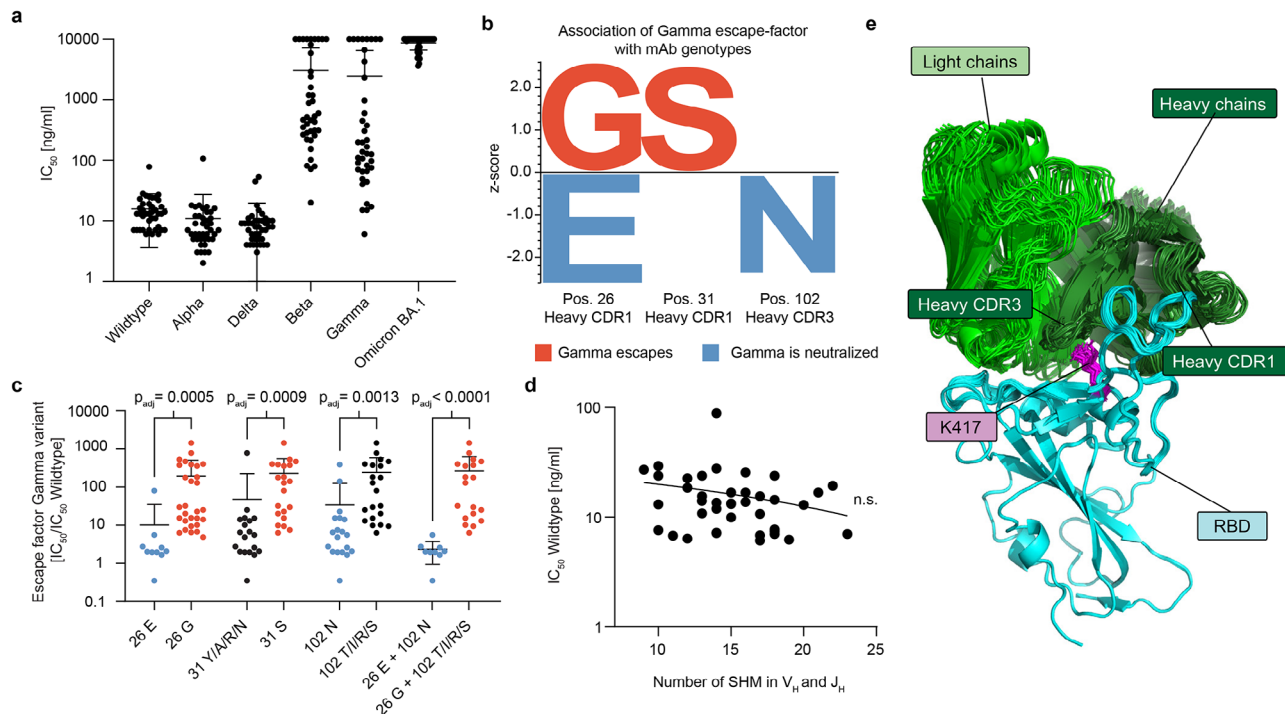


Figure 3. Specific SHM associate with a gain-of-neutralization phenotype. (a) Quantification of the experiment outlined in Fig. 2A. Nonneutralizing mAbs were set to an IC₅₀ of 10,000 ng/mL. Plotted is the mean $\pm$ SD, $n = 40$. (b) Amino acid residues that associate with cross-neutralization of the Gamma variant were identified via SigniSite. Higher z-scores associate with higher escape factors. A significance threshold of 0.05 was applied after the Bonferroni correction. (c) Stratification of the mAbs on their genotype as suggested in (b). Kruskal-Wallis test was performed with correction for multiple comparisons and adjusted P -values are indicated. Data represented as mean $\pm$ SD. (d) The IC₅₀ values were plotted against the divergence from the germline. (e) Molecular docking of some members of the clonotype was performed with P5A-1D2 (PDB 7CHO) as a reference and aligned using PyMol. The position of the heavy chain CDR3 and the K417 residue on the SARS-CoV-2 RBD is highlighted.

gene segments as the mAbs from clonotype 1 (Fig. S5). Docking simulations with all 40 unique antibodies from clonotype 1 indicated very similar binding poses for 25 of the mAbs, however, they were predicted to bind the antigen in a slightly shifted manner (Fig. 3e). The Beta and Gamma virus variants differed from the ancestral virus by the RBD mutations at residues N501, K417, and E484. Closer examination of the binding interface revealed the residue K417 of the RBD lies centrally in the epitope and in close proximity with the V_H CDR3 in all examined mAbs of clonotype 1 (Fig. 3e). This finding suggests that through certain SHM within the heavy chain CDR3 tolerance toward K417 substitutions was achieved, which would lead to a loss of affinity without these SHM.

SHM acquired by MBC during convalescence were previously shown to increase the neutralization potency and breadth of SARS-CoV-2-specific monoclonal antibodies when pairs of clonally related mAbs from very early and late time points after immunization were analyzed [18, 19]. However, it remained poorly understood how MBC are able to select for breadth-enhancing SHM without prior knowledge of the variants that are about to emerge in the future. Since many of the detected SHM did not account for any measurable effect, it is possible that MBC mature stochastically to diversify their receptor repertoire. By chance, some of the mutated BCR versions might fit new virus variants, which tend to escape from antibodies encoded by the most dom-

inant, unmutated germline sequences. The functional diversification of the BCR repertoire through SHM was similarly observed for B-cell responses against other viruses such as dengue and influenza [20, 21].

A comparable effect was recently observed in the context of SARS-CoV-2 neutralizing mAbs for the VH1-58 clonotype, in which bystander mutations mediate cross-neutralization of the Omicron variant, but not Beta and Gamma [22]. Our observation discovers the effect of SHM within the highly prevalent VH3-53 gene segment on these early VOCs and therefore completes our understanding of the impact of affinity maturation on SARS-CoV-2 VOC neutralization. In accordance with our findings, the establishment of an “anticipatory” MBC compartment was previously proposed by others [23].

An alternative model for the enhancement of neutralization breadth through repeated antigen contacts is the masking of immunodominant epitopes through pre-existing plasma antibodies [5, 24]. In our study, the infection-induced plasma antibody titer was close to undetectable, making an influence on the vaccine response very unlikely. Importantly, both concepts, (1) generation of an anticipatory memory response by stochastic SHM and (2) by masking of immunodominant epitopes, are not mutually exclusive and may be relevant in an additive manner, depending on the setting.

Conclusion

This study illuminates the importance of SHM for the diversification of expanded B-cell clonotypes using the VH3-53 immunoglobulin gene segment and that this process enhances the neutralization of VOC in a putatively stochastic manner. This may likely be an evolutionary advantageous function of affinity maturation to mediate neutralization of future virus variants that individuals have not (yet) been exposed to. A deeper understanding of which SHM are advantageous for the coverage of VOC will pave the way to the design of next-generation variant-adaptable recombinant mAb products for the treatment of infectious diseases.

Data limitations and perspectives

Our results were retrieved from one single blood donor and not from a representative cohort. This approach was necessary because expanded clonotypes are private and thus can only be found within the repertoire of single individuals. Our conclusion is in accordance with recently published data and therefore likely generalizable [22]. Our computational docking analysis is error-prone, and theoretically, the generated structures may deviate from experimentally determined ones. Nevertheless, in the special case of SARS-CoV-2, the availability of many solved antibody–antigen structures of antibodies with similar V_H and V_L sequences enhances the precision of prediction under the premise that two mAbs with strong homology show a similar binding pose.

Materials and methods

Study design

A blood donor, from whom three neutralizing antibodies were identified in a previous study [13], donated whole blood samples at days 0, 8, 16, and 71 after the first COVID-19 vaccination with Vaxcevia and after another 34 days following booster vaccination with Corminaty (administered at day 71). Importantly, both vaccines were not adapted to VOC. PBMC were isolated using Ficoll gradient centrifugation, aliquoted at 10^7 cells/mL, cryopreserved in FCS with 10% DMSO, and stored at -150°C .

ELISA and NeutralISA

Serial plasma dilutions were applied to the SARS-CoV-2 S1 IgG ELISA (Euroimmun EI 2606–9601 G) and the EC_{50} titer was determined by nonlinear regression analysis in GraphPad Prism 10.1.1. The NeutralISA assay (Euroimmun EI 2606-9601-4) was performed as described by the manufacturer's instructions and the OD_{450} signal was normalized to the pre-vaccination sample of the experiment.

Single-cell RNA and V(D)J sequencing of B cells derived from a vaccine response

Cryopreserved PBMC were thawed and 2×10^7 cells from the different time points were each labeled with 5 μg of anti-human antibodies carrying the hashtag sequences 1–5 (BioLegend 394661–69). The cells from the different time points were pooled, and SARS-CoV-2 RBD-specific B cells were identified using the SARS-CoV-2 RBD B Cell Analysis Kit (Miltenyi 130-128-032) under usage of the Viobility 405/520 Fixable Dye (Miltenyi 130-130-404), anti-CD19 VioBlue (Miltenyi 130-113-172) and anti-IgD APC (Miltenyi 130-099-221). Live $\text{CD}19^+$, IgD^- B cells were sorted using the Miltenyi Tyto sorter and analyzed using the Miltenyi MACSQuant Analyzer 10. The purified sample was processed with 10X Genomics scRNA-seq using the Next GEM Single Cell 5' Reagent Kit v2 (Protocol CG000330 Rev A) and sequenced using the Illumina NovaSeq 6000 (2×150 bp).

Bioinformatics analysis of scRNA-seq and V(D)J data

The raw multiplexed fastq files of scRNA-seq, V(D)J, and feature barcode sequences were processed using Cell Ranger v7.1.0. The 'multi' command was used for demultiplexing and aligning the scRNA and VDJ to human references GRCh38-2020-A and GRCh38-alt-ensembl-7.1.0, respectively. The filtered gene count matrices and fasta files of the V(D)J data were imported in Seurat [25] (v4.3.0) and Immcantation suite (v4.4.0), respectively in R (v4.2.1).

Transcripts that were expressed in <5 cells, cells in which <200 genes were expressed and cells with mitochondrial gene expression $>5\%$ were removed. Highly variable genes were selected via the 'vst' method, the data was scaled, and principal components analysis was performed with a default number ($n = 50$) of principal components (PC) in Seurat. The first 15 PC were selected for further clustering. Unsupervised clustering of the cells was performed by the "FindNeighbors" and "FindClusters" functions of Seurat using the default Louvain algorithm and a resolution value of 0.7. Non-B cells (T cells, NK cells, and monocytes) were removed. The data was reclustered with the first 10 PC and a resolution value of 0.2. Plots were generated via the unique manifold approximation and projection method. Monocle3 [26] (v1.3.1) was used for the trajectory and pseudotime analysis with the trajectory root being assigned within the infection-induced MBC cluster.

The V(D)J dataset was first analyzed via the Cell Ranger "vdj" command, and the clonal expansion and isotype diversity were visualized via the 10X Genomics enclone tool (v0.5.116). Further analysis was performed using the Change-O (v1.3.0) toolkit [27] with IgBLAST (v1.20.0) to perform V(D)J gene assignment by aligning the fasta files against the IMGT [28] human reference germline database. The number of somatic hypermutations was calculated for the V and J genes by multiplication of the germline divergence ($1 - \text{region identity to the IMGT germline reference}$) with the region length. For the identification of clonally related

sequences, the nearest-neighbor distance distribution method in SHazaM (1.1.2.999) was used for threshold calculation and clones were called via a hierarchical clustering approach (distance-based method) using SCOPer (v1.2.1.999). The clonal overlap between samples was visualized using circlize [29], with the pre-vaccination mAb MB027_D06L being incorporated as an additional sample. Lineage trees were constructed and plotted using the Dowser package (v1.1.1) and data were visualized using ggplot2 (v3.4.0).

Expression of fully human IgG1 monoclonal antibodies

Of selected antibody candidates, monoclonal antibodies (mAb) were expressed as previously described [13]. The variable regions, including the native signal peptide of the respective V_H or V_L gene segments, were gene-synthesized and cloned into pTRIOZ (InvivoGen ptrioz-higg1) for kappa or pTRIOZ-L2 for lambda light chains. 0.8 μ g/mL of the plasmids was transfected into Expi-CHO cells using the ExpiFectamine CHO Transfection Kit (Thermo Fisher A29129). The mAbs were purified using Protein G 24-well plates (Takara 635732) and dialyzed overnight in 100-fold PBS volume. Protein concentrations were determined by Qubit Protein Assay and the mAb purity and integrity were assessed by SDS-PAGE gel electrophoresis.

Neutralization breadth determination

The virus-neutralizing activity of the recombinant mAbs was determined for several VOC using the VSV pseudotype system as described previously [13, 30].

The different mAbs were threefold serially diluted in triplicates starting from 4 μ g/mL in a final volume of 75 μ L/well over 11 dilution steps. 300 ffu/well of the respective pseudotype virus was mixed with the mAb dilutions, incubated at 37°C for 1 h, transferred to Vero B4 cell monolayers, and incubated for 24 h. The infected cells that expressed GFP were identified by full-well wide-field fluorescence microscopy using an Olympus FV3000 microscope. The number of infected (GFP⁺) cells per well was quantified by ImageJ. Average values were calculated from the numbers of GFP⁺ cells in triplicates and normalized to control wells without mAb. The data were plotted using GraphPad Prism 10.1.1 with a nonlinear regression for curves that visibly show a neutralizing effect and with linear regression for curves with a lack of effect, and the IC_{50} values for each variant were calculated. An escape factor for each VOC was determined by normalizing the IC_{50} of the respective VOC on the IC_{50} of the wildtype variant. All neutralization experiments were validated in two independent experimental repeats. Association of the derived escape factor with residues in the primary amino acid sequence of the respective mAbs was performed using SigniSite (v2.1) [31].

Homology-controlled antibody–antigen docking

The protein sequences of the variable regions of mAbs were compared with all entries in the RCSB Protein Data Bank (PDB) using BLAST (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>). In order to perform antibody–antigen docking, only templates were chosen that shared at least 80% identity in the heavy and light chains, respectively. Tertiary structures of the variable regions were predicted using DeepAb [32] and renumbered using the re-numb function of the WHAT IF server at <https://swift.cmbi.umcn.nl>. Therefore, every amino acid coordinate in the antibody structures is respective to a virtual V(D)J-VJ fusion protein that starts with the first amino acid of the V_H and ends with the last amino acid of the J_L .

The renumbered files were docked on the SARS-CoV-2 RBD (PDB ID: 7CHO chain E [17]) using HADDOCK2.4 [33] (<https://wenmr.science.uu.nl/haddock2.4>) with specification of the CDR residues as active and the RBD surface as passive. On the antibody side, “automatically define passive residues around the active residues” was unchecked. On the RBD selection panel, the option “automatically define surface residues as passive” was selected. The HADDOCK sampling parameters were set to 10,000 structures for rigid body docking, 400 structures for semiflexible refinement, and 400 structures for final refinement. A total of 400 models were specified to be analyzed in the analysis parameter menu. The remaining settings were left on default. The derived pdb files were aligned and visualized via The PyMOL Molecular Graphics System, Version 2.5 Schrödinger, LLC. The method was previously utilized in a modified version [13].

Statistical analysis

The statistical analysis for linear and nonlinear regression as well as test for statistical significance between sample groups (Kruskal–Wallis test with correction for multiple hypotheses) were performed in GraphPad Prism 8.4.2. Association of mAb amino acid sequences with their phenotype was performed in SigniSite 2.1. Detailed statistical results and exact *P*-values are listed in Table S2.

Ethics approval and consent to participate

The blood donations were approved by the Ethics Committee of Hannover Medical School under application numbers 6259 and 10789_BO_S_2023. The donor gave written informed consent for the peripheral blood to be collected and used for research purposes at the Hannover Medical School (MHH).

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Conflict of interest: S.P. and M.H. conducted contract research (testing of vaccine sera for neutralizing activity against SARS-CoV-2) for Valneva unrelated to this work. S.P. further served as an advisor for BioNTech, unrelated to this work. The remaining authors declare no conflict of interest.

Author contributions: Conceptualization: Matthias Bruhn, Axel Schambach, and Ulrich Kalinke. Methodology: Matthias Bruhn, Maureen Obara, Bibiana Costa, Abdus Salam, Annett Ziegler, Inken Walzl, Andreas Pavlou, Markus Hoffmann, Theresa Graalmann, Stefan Pöhlmann, Axel Schambach, and Ulrich Kalinke. Investigation: Matthias Bruhn and Maureen Obara. Visualization: Matthias Bruhn and Abdus Salam. Funding acquisition: Matthias Bruhn, Stefan Pöhlmann, Axel Schambach, and Ulrich Kalinke. Project administration: Matthias Bruhn and Ulrich Kalinke. Supervision: Ulrich Kalinke. Writing – original draft: Matthias Bruhn and Ulrich Kalinke. Writing – review & editing: all authors.

Data availability statement: The sequences of all mAbs generated in this study are available for download in Table S1. The raw sequencing data including the libraries for gene expression, surface barcodes, and V(D)J profiling are available in the SRA database under the accession number PRJNA954694. The scripts for the analysis of the scRNA-seq experiment can be accessed under <https://github.com/memumab/Ant-Mem>.

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- Abbreviations:** **mAb**: monoclonal antibody · **BCR**: B-cell receptor · **CDR**: complementarity-determining region · **MBC**: memory B cell · **PC**: principal component · **PDB**: Protein Data Bank · **RBD**: receptor binding domain · **SARS-CoV-2**: severe acute respiratory syndrome coronavirus 2 · **SHM**: somatic hypermutation · **VOC**: variant of concern
- Full correspondence:** Prof. Ulrich Kalinke, Institute for Experimental Infection Research, TWINCORE, Centre for Experimental and Clinical Infection Research, Feodor-Lynen-Str. 7-9, Hannover 30625, Germany
e-mail: Ulrich.Kalinke@twincore.de
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